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2024 Taiwan clinical practice guideline for diabetic kidney disease – an executive summary

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ABSTRACT

Scientific advances and development in the management of diabetes including use of new glucose-lowering agents for cardiorenal protection in diabetic patients prompted revision of local clinical practice guideline for diabetic kidney disease (DKD). Multiple cross-disciplinary professional meetings were held in 2023 and 2024 by experts from the Diabetes Association of the Republic of China (Taiwan), the Taiwanese Association of Diabetes Educators and the Taiwan Society of Nephrology to review the latest evidence and to develop updated

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recommendations, taking local epidemiology, circumstances and relevant local guidelines into considerations. From screening and diagnosis; risk classification and monitoring; lifestyle modifications; glycemic, blood pressure and lipid management; to the use of cardiorenal protective medications and complication management, the 2024 Taiwan Clinical Practice Guideline for DKD aims to offer up-to-date reference and comprehensive guidance to local practitioners for optimization of DKD patient care.

1. Introduction

Since publication of Taiwan Clinical Practice Guideline for Diabetic Kidney Disease in 2019 [1], significant development has evolved in management of diabetic kidney disease (DKD) with emergence of new glucose-lowering agents demonstrating additional cardiorenal protective benefits [2,3]. The Diabetes Association of the Republic of China (Taiwan) (DAROC) collaborated with the Taiwanese Association of Diabetes Educators and the Taiwan Society of Nephrology to conduct meetings in 2023 and 2024 to review and discuss latest evidence, and to update recommendations for DKD patient management. This article provides an overview of key resolutions from this cross-disciplinary joint revision effort.

2. Epidemiology

Chronic kidney disease (CKD) is defined as renal structural or functional abnormalities (e.g. albuminuria or low estimated glomerular filtration rate, eGFR) persistent for at least 3 months [2–4]. DKD generally specifies CKD attributable to diabetes mellitus (DM) whereas recent Kidney Disease Improving Global Outcomes (KDIGO) guidelines highlighted the fact that traditional diabetes pathophysiology may not be the cause of CKD in all patients with diabetes mellitus (DM) [2,3]. Meanwhile, clinical approach to management of CKD and DM patients were practically similar thus KDIGO considered use of “DKD” in this regard appropriate as long as such constraint is acknowledged [3]. We hereby adopt the broader concept of DKD to include CKD-DM patients with other etiologies in this guideline. DKD can be classified by renal function trajectories into 4 types: (1) classical DKD; (2) albuminuria regression; (3) rapid eGFR decline; and (4) non-proteinuric or non-albuminuric DKD, each associated with different clinicopathological factors [5].

Amid global trend, national Nutrition and Health Survey in Taiwan revealed an ongoing increase of DM prevalence from 6.2 % during 1993–1996 to 7.8 % during 2005–2008, which further went up to 9.32 % in 2014 as estimated by the local Diabetes Atlas 2019 using data from national health insurance database [6–8]. A rising DKD prevalence from 9.17 % in 2000 to 17.92 % in 2014 was also recorded, pervasive across both sexes and all age groups but elderly were most affected [8]. Taiwan ranked the top on incidence of treated end-stage renal disease (ESRD)

globally (525 per million population in 2020), with DKD constituting 49.6 % of all dialysis-dependent patients reflecting its public health significance [9,10].

Hyperglycemia and hypertension were shown to be predictors of CKD development and progression [11,12]. A study in Taiwan also found hyperglycemia, hypertension and hyperlipidemia, together with age, race, family history, dietary patterns and lifestyles being risk factors for DKD development, indicating the importance of optimal glucose, blood pressure (BP) and lipid management in DKD care for improvement of prognosis [13].

3. Screening and diagnosis

For type 1 DM patients, DKD generally presents 10 years after diagnosis [2]. Although type 1 DM made up to <1 % of all DM patients in Taiwan, a registry identified DKD prevalence of 16.7 % with suboptimal use (around 20 % only) of angiotensin converting enzyme inhibitor (ACEi) and angiotensin receptor blocker (ARB) [14]. On the other hand, for type 2 DM patients, DKD is usually evident at the time of diagnosis, but the median testing rate of urine albumin-to-creatinine ratio (UACR) was merely around 50 % in the US [2,15]. Health technology assessment by the National Institute for Health and Care Excellence in the United Kingdom considered annual UACR testing as a cost-effective screening tool for early DKD in both type 1 and 2 DM patients, with type 2 being more cost-effective [16]. Besides recommendations by international guidelines as measurements for glomerular function and for diagnosis of DKD [2,3], changes in UACR and eGFR were both validated as appropriate trial end points for early stages of CKD including DKD [17]. As such, for all DM patients, we recommend annual screening for DKD by UACR and eGFR since diagnosis (Table 1). For UACR, spot urine samples (particularly collected in the morning) is preferred [2,4]. Reevaluation may be indicated as factors like infection, inflammation, heart failure, hyperglycemia, hypertension, menstruation or exercises can affect urine albumin excretion; and serum creatinine level thus eGFR may vary with muscle mass, age, sex, race, diet, medications and laboratory methods [1]. Referral to nephrologists and renal biopsy may be considered for exclusion of other etiologies in case of non-typical presentations [1,2].

Table 1

Recommendations on screening and diagnosis of DKD patients.

Recommendations	Class
All DM patients should undergo screening for DKD at least once annually since diagnosis.	Strong
Screening modalities should include UACR and serum creatinine for calculation of eGFR.	Strong
Urine albumin excretion can be influenced by various factors. Abnormalities detected by spot UACR should be reevaluated within a 3–6 months period.	Strong
Diagnosis of DKD can be made in DM patients with:	Strong
1 UACR ≥ 30 mg/g, OR	
2 eGFR < 60 mL/min/1.73m ² ,	
persistent for at least 3 months.	
In the presence of non-typical clinical presentations, alternative causes of kidney diseases should be excluded. Non-typical clinical presentations may include:	Strong
1 Albuminuria > 3.5 g/day within 5 years of DM diagnosis,	
2 Hematuria,	
3 Absence of diabetic retinopathy,	
4 Unexplained acute kidney injury,	
5 eGFR decline > 30 % after use of ACEi/ARB.	

ACEi, angiotensin converting enzyme inhibitor; ARB, angiotensin receptor blocker; DKD, diabetic kidney disease; DM, diabetes mellitus; eGFR, estimated glomerular filtration rate; UACR, urine albumin-to-creatinine ratio.

CKD is classified according to: - eGFR (G) - Albuminuria (A)				Albuminuria categories		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30 mg/g <3 mg/mmol	30-299 mg/g 3-29 mg/mmol	≥300 mg/g ≥30 mg/mmol
eGFR categories (mL/min/1.73m ²)	G1	Normal or high	≥90	Screen 1	Treat 1	Treat 3
	G2	Mildly decreased	60-89	Screen 1	Treat 1	Treat 3
	G3a	Mildly to moderately decreased	45-59	Treat 1	Treat 2	Treat 3
	G3b	Moderately to severely decreased	30-44	Treat 2	Treat 3	Treat 3
	G4	Severely decreased	15-29	Treat 3	Treat 3	Treat 4+
	G5	Renal failure	<15	Treat 4+	Treat 4+	Treat 4+
Low risk – for annual screening once by UACR and eGFR [^]						
Moderate risk – for annual testing once by UACR and eGFR [^]						
High risk – for annual testing 2-3 times by UACR and eGFR [^]						
Very high risk – for annual screening 3 times by UACR and eGFR [^]						
Very high risk – for annual screening at least 4 times by UACR and eGFR [^]						

Fig. 1. Recommended risk classification of DKD patients (adapted from Refs. [4,19]). CKD: chronic kidney disease; DKD: diabetic kidney disease; eGFR: estimated glomerular filtration rate; UACR, urine albumin-to-creatinine ratio. [^] Diabetes Shared Care Network Program in Taiwan requires enrolled DKD patients to have annual and periodic (3-monthly) assessments with provision of associated financial incentives to the healthcare institutions [20]. Program participants were shown to have lower hemodialysis rate, lower hospitalization events and higher overall survival thus enrollment should be encouraged [21].

4. Risk classification and monitoring

UACR levels were predictors of CKD progression as well as cardiovascular and renal outcomes for DM patients [18]. According to levels of UACR and eGFR, DKD patients are classified by risk of CKD progression to guide frequency of monitoring, treatment decision and need for referral to nephrologists (Fig. 1, adapted from Refs. [4,19]). All DM patients should have UACR, eGFR and blood potassium level ± other renal function related electrolytes, HCO₃, calcium, phosphorus, parathyroid hormone, hemoglobin and serum albumin monitored at least once annually depending on eGFR levels and other individual conditions [1]. In Taiwan, the government-led Pay-for-Performance (P4P) Program (Diabetes Shared Care Network Program) requires enrolled DKD patients to have annual and periodic (3-monthly) assessments of various quality of care indicators including UACR and eGFR, with provision of associated financial incentives to the healthcare institutions [20]. P4P program participants were shown to have lower hemodialysis rate,

lower hospitalization events and higher overall survival therefore enrollment should be encouraged [21].

We recommend patients with the following conditions to be referred to nephrologists for assessment and management [1,2]:

- 1) continuously rising UACR levels,
- 2) continuously declining eGFR levels,
- 3) difficulty in management of primary diseases,
- 4) uncertain etiology of kidney disease,
- 5) eGFR <30 mL/min/1.73 m², and
- 6) need for discussion of renal replacement therapy.

Besides, referral may also be considered for patients at G3b (i.e. eGFR <45 mL/min/1.73 m²) [1].

5. Nutrition, exercise and lifestyle

DKD patients can benefit from holistic care including nutrition, exercise, weight management, smoking cessation, medical therapy and control of comorbidities, emphasizing patient-centered and multidisciplinary approaches [2,22]. Diabetes self-management education and support (DSMES) facilities knowledge, skill and ability required for implementing and sustaining DM self-care, and was shown to improve health literacy, quality of care and clinical outcomes [22–24]. Chronic care model (CCM) is a frequently adopted framework for enhancement of care for patients with chronic diseases and was demonstrated to delay disease progression and prevent complications in DM patients [25,26].

Medical nutrition therapy provided by dietitians or nutritionists can offer benefits by optimizing nutritional status and minimizing risks associated with comorbidities in CKD patients; and was demonstrated to slow disease progression and time to dialysis [27,28]. We therefore recommend regular nutritional assessment for all DKD patients. DKD patients should have a balanced intake of vegetables, fruits, whole grains, proteins, unsaturated fatty acids and nuts, as well as reduced intake of sugar-sweetened beverages and refined carbohydrates [1,2]. Dietary Approaches to Stop Hypertension (DASH) and Mediterranean diet may reduce cardiovascular risk by improving glycemic control [29]. Plant-based protein and high-fiber diets were associated with lower risk of all-cause and/or cardiovascular mortality [30,31]. However, for late stage DKD patients, attention should be paid to electrolytes and adjustment according to clinical conditions may be required. A wide range of disorders may develop as a consequence of the loss of kidney function. These include disorders of fluid and electrolyte balance, such as volume overload, hyperkalemia, and hyperphosphatemia. Individualized monitoring of key electrolytes (e.g., potassium, phosphorus, sodium) is strongly advised. People with late stage DKD should receive appropriate diet instructions to limit the intake of foods rich in bioavailable potassium (e.g., processed foods). The specific nutritional intervention recommends daily sodium intake <2 g (or sodium chloride <5 g/d), potassium intake around 2–4 g per day, and phosphate restriction as 5–10 mg/kg/d [21].

Calorie intake of DKD patients should be similar to the general population, recommended as 25–35 kcal/kg/day, and can be adjusted according to individual level of activity [27]. For non-dialysis dependent DKD patients, daily protein intake should be maintained at around 0.8 g/kg, with plant-based and low-fat animal-based proteins as major sources [2,32] because there is no conclusive evidence that restriction of dietary protein to levels <0.8 g/kg/day improves kidney or other health outcomes [19]. Further restriction is not recommended as no additional benefit was shown on renal and other outcomes [33]. Although there has not been a full elucidation of the underlying mechanisms by which high

protein intake may adversely affect kidney function, existing data suggest that glomerular hyperfiltration caused by a high-protein diet may lead to an increase in albuminuria and an initial rise and subsequent decline in GFR. Moreover, growing evidence suggests that high-protein diets may be associated with a number of metabolic complications that may be detrimental to kidney health [3,21]. However, all recommendations call for higher levels of protein intake for patients with kidney failure treated with maintenance dialysis, who are often catabolic or malnourished (e.g., 1.0–1.2 g/kg/day) [19].

Sodium intake reduction was found to benefit BP control and lower cardiovascular risks [34,35]. For DKD patients, daily sodium intake is recommended to be maintained at below 2000–2300 mg for better BP control and to reduce albuminuria [2,3].

DKD patients should maintain an optimal body weight. Overweight or obese patients should reach an energy deficit of 500–750 kcal/day through nutrition and exercise [36]. Reduction in body weight of 5–10 % and above may slow progression of renal disease [37]. We recommend DKD patients to exercise regularly, including moderate-intensity physical activity at least 150 min/week or at an equivalent level compatible with cardiovascular and physical tolerance [3].

Tobacco use and secondhand smoke were known risk factors associated with CKD development [38,39]. Evidence also suggested potential harm to respiratory and cardiovascular systems by newer forms of cigarettes [40]. DKD patients should avoid use of any tobacco products (including traditional and new forms) and avoid secondhand smoke exposure. Smokers should quit smoking immediately.

In Taiwan, betel-quid chewing prevalence in the general population was decreasing over the years although the rate remained high at around 10 % among men of 40–49 years in 2018 [41]. Given betel-quid chewing was identified as independent risk factors for CKD and proteinuria, as well as being associated with hypertension and metabolic syndrome, DKD patients should avoid betel quid chewing and those with regular chewing habits should quit immediately [42–47].

While heavy drinking was associated with harm of CKD development, moderate consumption was not and appeared to have potential protective effects instead [48,49]. However, alcohol consumption in DKD patients with pre-existing liver disease is strongly not recommended. Patients with hepatic impairment are at significantly increased risk of liver damage from alcohol, and this can further complicate their overall health status and DKD management. Additionally, alcohol can interfere with uric acid excretion, potentially leading to or worsening hyperuricemia and gout, which can have negative consequences for kidney function and cardiovascular health. Even for DKD patients with good glycemic control and without contraindications or other special considerations, alcohol consumption should be restricted to less than 2 standard drinks for males and 1 standard drink for females per day [50]. Drinking on an empty stomach should be avoided.

6. Glucose, blood pressure and lipid management

6.1. Glucose control

Optimal glycemic control may reduce or slow occurrence of albuminuria and renal function decline in DKD patients as shown by overseas and local studies [51–53]. However, glycemic targets should be individualized according to age, comorbidities, cognitive function etc. of individual DM patients to prevent hypoglycemic events for balance of benefits and harms (Supplementary Table 1) [2,3,54]. For patients with glycemic targets stably met, we recommend monitoring of glycemic indicators (e.g. glycated hemoglobin, HbA1c) at least twice annually. For patients not meeting glycemic targets or with recent adjustment of glucose-lowering medications, we recommend monitoring of glycemic indicators at least every 3 months and increase in frequency of

monitoring according to clinical condition [3,54]. For DKD patients, accuracy of HbA1c has its limitations. Other indicators such as glycated albumin (GA), self-monitoring blood glucose (SMBG), continuous glucose monitoring (CGM), fasting and post-prandial monitoring etc. may be considered [3,54].

General principles of glucose-lowering medical therapy for local DM patients are detailed in the Clinical Practice Guidelines for Type 2 Diabetes Care 2022 published by DAROC [55]. Characteristics of various medications with considerations for use and dose adjustment required are highlighted in Table 2. Priority should be given to agents with additional organ protective benefits independent of glycemic level [1].

6.2. Blood pressure control

Hypertension, defined by the average of two or more BP measurements at two or more occasions with systolic pressure ≥ 130 mmHg or diastolic pressure ≥ 80 mmHg, is an important predictor of CVD [56]. We recommend DKD patients to have BP measured during every clinic visit. Home BP monitoring is recommended for screening of hypertension as well as for regular monitoring once hypertension is diagnosed, and guidance should be provided to patients for standardized measurements [35,57]. Most current guidelines and consensus suggest the “722” protocol is optimal for the diagnosis and management of hypertension: for consecutive “7” (≥ 4) days, at “2” occasions (morning and evening) and with “ ≥ 2 ” (≥ 3 if atrial fibrillation) BP readings 1 min apart on each occasion [57]. For patients with elevated BP without a diagnosis of hypertension (i.e. systolic pressure 120–129 mmHg and diastolic pressure < 80 mmHg), multiple measurements should be taken [56]. DKD patients are at higher cardiovascular risks than other hypertensive patients. Lifestyle modifications including appropriate nutrition (e.g. DASH diet) and exercise should be recommended as aforementioned for all patients with elevated blood pressure $> 120/80$ mmHg [56–58].

BP control targets of DKD patients with hypertension should adopt a shared decision-making approach, with comprehensive consideration of risks of CVD, adverse events of medications, patient preference etc. [56]. We recommend a BP control target of $< 130/80$ mmHg and may consider further restriction to below 120 mmHg depending on patient’s tolerance [55]. If BP persists at $\geq 130/80$ mmHg, we recommend initiation of medication in addition to lifestyle modifications. For home BP $\geq 20/10$ mmHg above target values, immediate administration of two antihypertensive drugs from different classes is recommended (Fig. 2) [57,59].

ACEi, ARB, dihydropyridine calcium channel blockers (DHP CCB) and thiazide-like diuretics are all choices of monotherapy for DKD patients with evidence to reduce cardiovascular events in DM patients [59, 60]. However, for DM patients with albuminuria, ACEi or ARB use can slow renal disease progression [60–63]. As such, use of an ACEi or ARB at maximum tolerated dose as first-line therapy for BP control for DKD patients with UACR ≥ 300 mg/g is strongly recommended. For DKD patients with $300 \text{ mg/g} > \text{UACR} \geq 30 \text{ mg/g}$, ACEi or ARB at maximum tolerated dose as first-line therapy is also recommended. For all DKD patients, regardless of presence of albuminuria, ACEi or ARB can be used as first-line therapy. Replacement by another ACEi or ARB should be attempted in case of intolerance because achieving optimal BP control often requires a more nuanced approach than simply increasing the dose or switching between ACEi/ARBs in cases of suboptimal response. If the BP target is not reached, and especially in the presence of hypertension-mediated organ damage (HMOD), add-on antihypertensive agents from other drug classes with complementary mechanisms of action should be considered. Considering agents with longer half-lives for smoother BP control throughout the day, agents with higher efficacy in lowering BP can be utilized, if appropriate.

For pregnant women with mild chronic hypertension, the Chronic Hypertension and Pregnancy (CHAP) trial showed that controlling BP at

<140/90 mmHg was associated with better pregnancy outcomes (significantly lower incidence of preeclampsia and preterm birth) [64]. We therefore recommend a baseline target of 140/90 mmHg for BP control in pregnant women. Further control to 110-135/85 mmHg may reduce risk of accelerated maternal hypertension yet antihypertensive use should be reduced or stopped for BP < 90/60 mmHg [56]. For women of reproductive age, medications with potential harm to pregnant women such as ACEi, ARB, statin etc. should be used cautiously and should be discontinued prior to conception.

6.3. Lipid management

Hyperlipidemia, DM and CKD are all important predictors of CVD with many shared risk factors [65,66]. Blood lipids should thus be screened for all DKD patients. For DKD patients receiving medications for lipid management, we recommend monitoring of blood lipids before treatment initiation, during dose adjustment, 4–12 weeks after treatment initiation/adjustment and at least annually afterwards [56].

Weight loss and smoking cessation are crucial for prevention and

control of CVD for patients with hyperlipidemia, while diet and physical activity interventions were shown to have beneficial effects on lipids [31]. For patients with hypertriglyceridemia (≥ 150 mg/dL [1.7 mmol/L]) or low high density lipoprotein cholesterol (HDL-C) (male <40 mg/dL [1.0 mmol/L]; female <50 mg/dL [1.3 mmol/L]), intensified lifestyle modification and glycemic control is recommended [56].

Given risk reduction of major CVD outcomes by lowering low density lipoprotein cholesterol (LDL-C) with statin was demonstrated among CKD and DM patients, use of statin for primary and secondary prevention is recommended for DKD patients with hyperlipidemia besides lifestyle modification [67,68]. For DKD patients, use of high intensity statin to reduce ≥ 50 % LDL-C is recommended with primary and secondary prevention targets of <70 mg/dL and <55 mg/dL respectively. When target is not met, addition of ezetimibe or PCSK9 inhibitor may be considered. For patients >75 years, statin treatment may be considered after discussion of potential benefits and risks and continuation is reasonable for those already on statin [69]. Systematic reviews and meta-analyses did not reveal significantly increased risk of rhabdomyolysis and liver function derangement for monotherapy of statin or

Table 2

Characteristics, considerations for use and dose adjustment required for glucose-lowering medications in DKD patients.

<i>Oral agents</i>					
Medications	Characteristics & considerations	eGFR (mL/min/1.73m ²)			
		≥ 60	$\geq 30 - <60$	$\geq 15 - <30$	<15
Biguanide - <i>Metformin</i>	- Dose adjustment required for prevention of lactic acidosis. - Not recommended for type 2 DM patients >80 years without prior use. - Not associated with weight gain. Hypoglycemia less common as monotherapy. - Gastrointestinal side effects possible. - Vitamin B₁₂ deficiency assessment recommended for those with >4 years use.		For eGFR $\geq 30-45$: not exceeding 1000mg daily		
Insulin secretagogues - <i>Sulfonylureas</i> Gliclazide Glimepiride Glipizide	- All insulin secretagogues have largely similar hypoglycemic effects. - May lead to hypoglycemia or weight gain. - Shorter acting agents (e.g. non-sulfonylureas) should be used for those with higher risks of hypoglycemia (e.g. elderly, liver/renal function impairment or higher glycemic fluctuations) if indicated.		Should use with caution for increased risk of hypoglycemia		
Glibenclamide	- Non-sulfonylureas (glinides: repaglinide, nateglinide and mitiglinide) mostly used to				

(continued on next page)

Table 2 (continued)

	reduce post-prandial hyperglycemia. - Combination use of sulfonylureas and glinides not recommended.			
α-glucosidase inhibitor -Acarbose -Miglitol	- Can reduce post-prandial hyperglycemia. - May have gastrointestinal side effects. - Not associated with weight gain. No hypoglycemia as monotherapy. - In case of hypoglycemia, treatment with monosaccharide (e.g. glucose) is recommended.			For eGFR<25: not recommended for use.
Thiazolidinedione -Pioglitazone	- Not recommended for use in patients with liver function derangement (ALT 2.5 times of ULN) or NYHA heart failure class III/IV. - Takes around 6-12 weeks for maximal effects. - Fluid retention, edema and weight gain may occur. - May increase risk of edema and congestive heart failure if combined use with insulin.			
Dipeptidyl peptidase 4 inhibitors (DPP4i) - Sitagliptin - Saxagliptin - Vildagliptin - Linagliptin - Alogliptin	- Side effects may include mild infections such as nasopharyngitis, and acute pancreatitis. - Not associated with weight gain. Hypoglycemia less common as monotherapy. - Dose adjustment required except for linagliptin.		For eGFR ≥ 30 –45: 50mg daily	25mg daily
			For eGFR <45: 2.5mg daily	
			For eGFR <50: 50mg daily	
			12.5mg daily	6.25mg daily

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combination use with ezetimibe, including in patients with CKD [70, 71]. For patients using statin, we recommend careful monitoring on clinical use with maximum tolerated dose. For advanced DKD patients with CVD events already on statin, continuation of use may be considered. However, for DKD patients entering stage of dialysis, initiation of statin treatment is not recommended [69].

For patients with hypertriglyceridemia, comprehensive assessment should be made on lifestyle factors (e.g. obesity or metabolic syndrome) and potential secondary causes (DM, chronic liver and renal diseases, nephrotic syndrome, hypothyroidism, effects from medications etc.) [72]. Diabetic dyslipidemia involves small, dense, atherogenic LDL particles, high triglyceride, and low HDL-C levels. Elevated triglyceride levels are associated with residual lipoprotein particles and an increase

in small, dense LDL particles [73]. For elevated triglyceride levels of ≥ 1000 mg/dL, fibrate medication should be considered to reduce risk of pancreatitis [56,69,72]. Renal function should be noted during use of fenofibrates. Dose reduction is recommended for $15 \leq \text{eGFR} < 60$ mL/min/1.73 m². For eGFR <15 mL/min/1.73 m², use should be avoided (Table 3). For DKD patients with elevated triglyceride levels (135–499 mg/dL) but LDL-C meeting target level, addition of icosapent ethyl can reduce risks of CVD [74]. Statin-fibrate and statin-niacin combinations were not shown to provide additional cardiovascular benefits (with the latter having potential of increased risks of stroke and side effects), combination use is therefore not recommended [75,76]. Moreover, there is no evidence to date to suggest lipid-lowering treatment would have a negative impact on cognitive function [77].

Table 2 (continued)

Sodium-glucose cotransporter 2 inhibitors (SGLT2i)	- Hypoglycemia less common. Usually associated with weight and blood pressure reduction.		100mg daily	Careful assessment for necessity of use
-Canagliflozin				
-Dapagliflozin	- May increase risk of urinary and genital tract infections. - Can reduce renal disease progression and risk of heart failure hospitalization. - For DKD patients with eGFR ≥ 20 mL/min/1.73m ² ,			For eGFR ≥ 25 : may initiate use and continue to use.
				For eGFR < 25 , initiation not recommended.
-Empagliflozin	recommend first-line combination use of metformin and SGLT2i in addition to lifestyle modification for glycemic control. - For patients with eGFR < 20 mL/min/1.73m ² already using SGLT2i, continuation is			10mg: may initiate use and continue to use. Not recommended for initiation for patients on dialysis.
				25mg: not recommended for use
-Ertugliflozin	recommended unless patient is intolerant or starts dialysis. - In case of volume depletion, should first consider reduction of diuretic use and education on fluid intake before considering initiation of SGLT2i. - Uncertainty remains regarding recommendations for post renal transplant patients.			For eGFR < 45 : not recommended for use.
Glucagon-like peptide-1 (GLP-1) receptor agonists (RA)	- With gastrointestinal side effects. - Significant weight reduction.			

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7. Cardiorenal protective medications

As recommended in our last published guideline, priority should be given to glucose-lowering agents with additional organ protective benefits independent of glycemic level [1]. These agents include sodium-glucose cotransporter 2 inhibitors (SGLT2i), glucagon-like peptide-1 (GLP-1) receptor agonists (RA) and non-steroidal mineralocorticoid receptor antagonist (nsMRA) (Table 4).

7.1. Sodium-glucose cotransporter 2 inhibitors (SGLT2i)

In the EMPA-REG OUTCOME trial, empagliflozin was associated

with risk reductions of 35 % for hospitalization for heart failure, 44 % for doubling of serum creatinine level and 55 % for renal replacement therapy in DM patients [78,79]. A more recent trial on CKD patients demonstrated a similar effect of risk reduction for composite outcome of renal disease progression or death from cardiovascular causes, although risk reductions for heart failure hospitalization or cardiovascular deaths were not statistically significant [80]. For dapagliflozin, DAPA-CKD trial was stopped early due to evident efficacy – a significant reduction (hazard ratio 0.61) of primary outcome event (sustained decline in eGFR ≥ 50 %, ESRD or deaths from renal/CVD causes) in both patients with and without type 2 DM [81]. Dapagliflozin was also shown to reduce risk of cardiovascular death or hospitalization for heart failure in

Table 2 (continued)

-Semaglutide	- Can reduce DM associated risks of overall and cardiovascular mortalities. - May consider use of GLP-1RA to reduce albuminuria and for renal protection.				
Injectable agents					
Medications	Characteristics & considerations	eGFR (mL/min/1.73m ²)			
		≥60	≥30 - <60	≥15 - <30	<15
Insulin	- May lead to hypoglycemia or weight gain. - For newly diagnosed type 2 DM patients with significant symptoms, and/or HbA1c ≥10%, and/or blood glucose ≥300 mg/dL, consider insulin as initiating treatment. - For type 2 DM patients with symptomatic hyperglycemia and HbA1c persists ≥7.5% or ≥1.5% above individualized target after non-pharmacological or non-insulin therapy, insulin therapy is recommended. - For type 2 DM patients not meeting glycemic target, insulin therapy should not be delayed. - Avoid excessive basal insulin dose (≥0.5 U/kg).	- Increase frequency of glycemic monitoring. Alert for hypoglycemia. Adjust dose if indicated. - For eGFR<10: consider dose reduction. Closely monitor glycemic level to prevent hypoglycemia			
Glucagon-like peptide-1 (GLP-1) receptor agonists (RA)	- With gastrointestinal side effects. - Significant weight reduction.	- For CCr 30-50 mL/min: should initiate from low dose and cautiously increase dose. - For CCr <30 mL/min: contraindicated. - For post-transplant: use with caution.			
-Exenatide	- Can reduce DM associated				

(continued on next page)

patients with heart failure and reduced ejection fraction, irrespective of DM status [82]. In Taiwan, dapagliflozin was approved for indications of prevention of CKD emergence or progression in type 2 DM patients, as well as for management of CKD and heart failure [83]. Similarly, canagliflozin was shown to lower risk of renal failure and cardiovascular events in CANVAS, CANVAS-R and CREDENCE trials [84,85]. Such cardiorenal protective benefits of SGLT2i were observed in patients with eGFR <45 mL/min/1.73 m² [81,84,85], including patients with eGFR ≥25 mL/min/1.73 m² in DAPA-CKD trial and eGFR ≥20 mL/min/1.73 m² in EMPEROR trial [81,86]. As such, KDIGO guideline had lowered the threshold for SGLT2i initiation accordingly and considered reasonable for continuation of use even if eGFR falls below <20 mL/min/1.73 m² unless intolerant [4].

Acute kidney injury (AKI) due to volume depletion was a potential concern, however disputed by a systematic review and meta-analysis

showing SGLT2i use was associated with lower risk of AKI [2,87]. For some patients, assessment of risk of hypoglycemia and volume depletion (e.g. concurrent diuretic use, history of AKI) may be practical on initiation of SGLT2i [3,88]. Risk of mycotic genital tract infections exists, which are usually mild and managed by topical treatment [4].

7.2. Glucagon-like peptide-1 (GLP-1) receptor agonists (RA)

Efficacy of GLP-1 RA was shown by the 2016 LEADER, 2017 EXSCEL and other trials in subsequent years [89,90]. A systematic review and meta-analysis found that GLP-1 RA reduced all-cause mortality, hospital admission for heart failure and composite kidney outcome (including macroalbuminuria, serum creatinine doubling, ≥40 % eGFR decline etc.) without increased in risk of severe hypoglycemia, retinopathy or pancreatic adverse events [91]. Recently, the FLOW (Evaluate Renal

Table 2 (continued)

-Exenatide extended release	risks of overall and cardiovascular mortalities. - May consider use of GLP-1RA to reduce albuminuria.	- For CCr 30-50 mL/min: should use with caution. - For CCr <30 mL/min or ESRD: contraindicated. - For post-transplant: use with caution.		
-Liraglutide		- No adjustment required. - According to TFDA approval, safety evaluation for eGFR 30-60 included 26-week assessment in clinical trials. - Canadian guideline does not recommend use for eGFR<50.		
-Semaglutide				
-Albiglutide				
-Dulaglutide				
-Lixisenatide			Use with caution and increase frequency of monitoring for renal function and side effects.	

No dose adjustment required
 Use with caution
 Not recommended

Function with Semaglutide Once Weekly) trial was the first dedicated kidney outcomes trial in DKD. A clear benefit of treatment with semaglutide on kidney outcomes was shown as well as on CV outcomes and survival compared with placebo [92]. Therefore, GLP-1 RAs may become another pillar of therapy for DKD along with ACEi/ARB, nsMRA and SGLT2i. Similarly, the results of Soul (Oral Semaglutide and Cardiovascular Outcomes in High-Risk Type 2 Diabetes) trial show a cardiovascular benefit of oral semaglutide and are consistent with results reported for injectable semaglutide and other GLP-1 RAs with demonstrated cardiovascular efficacy [93].

Nausea, diarrhea, vomiting and other gastrointestinal symptoms are common adverse events associated with GLP1-RA [94–96]. Liraglutide is not recommended for use in patients with creatinine clearance <30 mL/min [94]. On the other hand, semaglutide and dulaglutide can be used in patients with renal impairment (including ESRD) with no dose adjustment required [95,96].

7.3. Non-steroidal mineralocorticoid receptor antagonist (nsMRA)

NsMRA Finerenone was shown to reduce risk of CKD progression, cardiovascular events and new-onset heart failure in DKD patients in FIDELIO-DKD and FIGARO-DKD trials [97–99]. Pooled analysis of these two trials concluded that addition of finerenone on top of maximum tolerated ACEi/ARB can improve CVD outcomes and slow CKD progression [99]. Of note, potassium level should be regularly monitored for risk of hyperkalemia and finerenone is not recommended for patients with eGFR <25 mL/min/1.73 m² [100].

8. Complication education and management

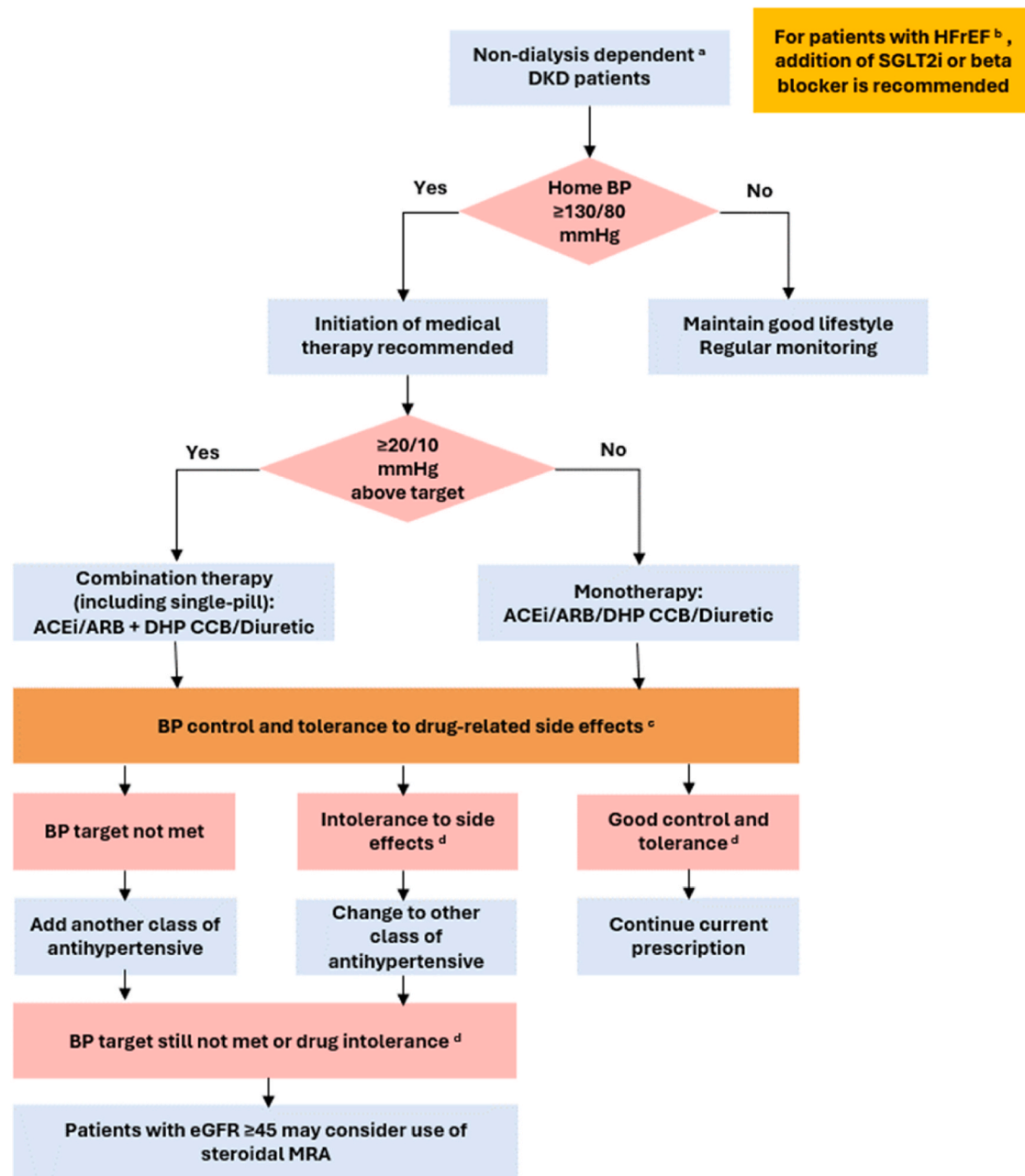
Multidisciplinary patient care and education to promote awareness, optimally in alignment with government-led shared care network, is recommended for DKD patients in Taiwan to reduce risk of complications and to improve clinical outcomes [21]. We recommend regular assessments for appropriate management of complications including hypoglycemic events, CVD (including coronary artery disease), neuropathy, retinopathy and diabetic foot for all DKD patients (Supplementary Table 2).

9. Conclusion

The 2024 Taiwan Clinical Practice Guideline reviewed the latest epidemiology and emerging evidence on management of DKD patients, including use of new glucose-lowering agents for additional cardiorenal protective benefits. Updated recommendations regarding screening and diagnosis, risk classification and monitoring, lifestyle (including nutrition and exercise), glycemic and BP targets, lipid control, use of cardiorenal protective medications, complication education and management were developed to provide guidance and reference for local healthcare workers for enhancement of quality care for DKD patients in Taiwan.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.



ACEi, angiotensin converting enzyme inhibitor; ARB, angiotensin receptor blocker; BP, blood pressure; DHP CCB: dihydropyridine calcium channel blocker; eGFR: estimated glomerular filtration rate; HFrEF: heart failure with reduced ejection function; MRA: mineralocorticoid receptor antagonists; SGLT2i: sodium-glucose cotransporter 2 inhibitors

^a For dialysis patients, blood pressure (BP) monitoring should be based on interdialytic home BP or ambulatory BP monitoring as compared to pre- and post-dialytic BP and should consider a control target of <130/80 mmHg [57].

^b For patients with heart failure with reduced ejection function (HFrEF), addition of sodium-glucose cotransporter 2 inhibitors (SGLT2i) or beta blocker can reduce mortality besides BP control [57].

^c For patients without albuminuria, ACEi/ARB/DHP CCB/Diuretic are all choices of monotherapy [56,57]. For patients with albuminuria, priority should be given to ACEi/ARB [60-63].

^d BP monitoring should follow the "722" protocol [57]. Use of ACEi/ARB at maximum tolerated dose and monitoring of serum creatinine and potassium level within 2 weeks after initiation is recommended [4,56].

Fig. 2. Medical therapy for hypertension in DKD patients [4,56,57,59-63].

Table 3

Dose adjustment required for lipid-lowering medications in DKD patients.

Medications	eGFR (mL/min/1.73m ²)			
	≥60	≥30 - <60	≥15 - <30	<15
Atorvastatin	No dose adjustment required			
Rosuvastatin	No dose adjustment required		Start from 5 mg/day Maximum dose 10 mg/day	
Pravastatin	No dose adjustment required			
Fluvastatin	No dose adjustment required	Uncertainty about evidence		
Simvastatin	No dose adjustment required		Use with caution from 5 mg/day	
Pitavastatin	No dose adjustment required	Start from 1 mg/day Maximum dose 2 mg/day		
Ezetimibe	No dose adjustment required			
Fenofibrate	No dose adjustment required	For dose reduction	Use contraindicated	
Gemfibrozil	No dose adjustment required			
Omega-3 fatty acids	No dose adjustment required			

eGFR: estimated glomerular filtration rate.

Table 4

Recommendations on use of cardiorenal protective medications in DKD patients.

Recommendations	Class
For DKD patients with eGFR ≥20 mL/min/1.73m ² , we recommend use of SGLT2i to reduce the occurrence of cardiovascular events and renal disease progression.	Strong
For DKD patients, GLP-1 RA is recommended to reduce occurrence of cardiovascular events and renal disease progression.	Strong
For DKD patients especially those with albuminuria, use of nsMRA in addition to maximum tolerated dose of ACEi or ARB can lower cardiovascular risks and slow renal disease progression.	Strong

ACEi, angiotensin converting enzyme inhibitor; ARB, angiotensin receptor blocker; DKD: diabetic kidney disease; eGFR: estimated glomerular filtration rate; ESRD: end stage renal disease; GLP-1 RA: Glucagon-like peptide-1 receptor agonist; nsMRA: non-steroidal mineralocorticoid receptor antagonist; SGLT2i: sodium-glucose cotransporter 2 inhibitors; UACR, urine albumin-to-creatinine ratio.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jfma.2025.08.020>.

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